

SRI LANKAN SIGN LANGUAGE

Sri Lankan Sign Language (SLSL) is a visual-gestural language used by the deaf community in Sri Lanka. It enables communication through hand gestures, facial expressions, and body movements. The language contains more than **2000 commonly used signs**, including alphabets, numbers, and everyday word gestures. Similar to spoken languages, SLSL has its own linguistic structure and grammar.

Main Linguistic Parameters of Sri Lankan Sign Language

1. Handshape
2. Orientation
3. Location
4. Movement
5. Non-manual signals

Handshape	Orientation-	Location	Movement	Non-Manual Signals
This refers to the shape of the fingers and hand when making a sign.	Orientation describes the direction the palm or fingers face	Location refers to where the sign is performed relative to the body.	Movement refers to the motion of the hands while signing. Movement helps differentiate dynamic signs.	These are gestures not made by the hands. These signals add meaning to the sign.
• number of fingers extended • finger position • finger bending	• inward • outward • upward • downward	• head area • face • chest • shoulder area	• circular motion • up/down motion • forward/backward motion	• facial expressions • mouth gestures • body posture • head movement

➤ Types of Signs in Sinhala Sign Language

1. Conversational Signs

Conversational signs represent **complete words or phrases** used in everyday communication. These gestures usually have a **direct correspondence with Sinhala words**.

Examples:

- greeting signs
- common daily words
- action signs

These signs usually have one-to-one correspondence with Sinhala words.

2. Fingerspelling Signs

Fingerspelling is used when:

- a word does not have a sign gesture
- proper names must be expressed
- new words appear

Each alphabet letter is represented by a sign.

Example:

Name “Sumith”

S → U → M → I → TH

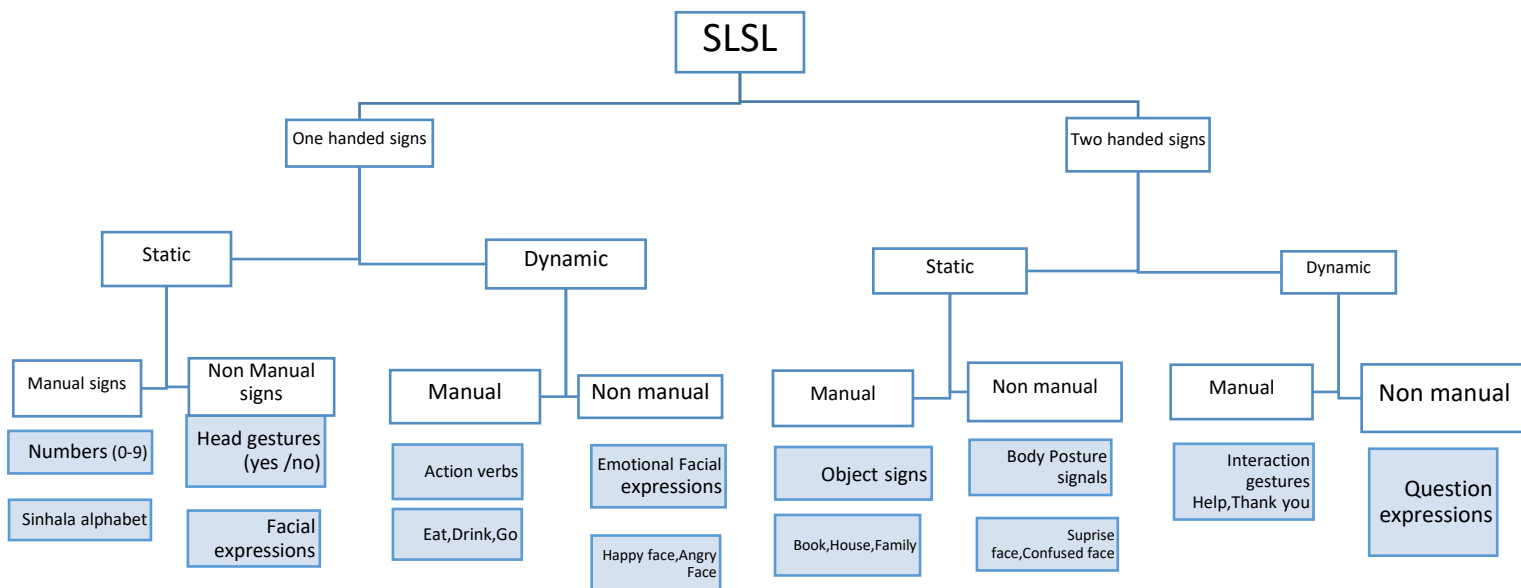
Sri Lankan Sign Language Gesture Categorization

Static Gestures

Static gestures involve **no significant movement of the hand**. The meaning is determined by the handshape and orientation.

Dynamic Gestures

Dynamic gestures involve **hand movement** and are commonly used to represent actions or expressions.



Challenges in Sign Language Recognition

1. Similar Handshape Gestures

Many signs use the **same or very similar handshapes**, but their meanings change depending on movement, orientation, or location. This makes it difficult for recognition systems to correctly distinguish between gestures.

2. Movement Variations

Some signs differ only by **hand movement direction or trajectory**. For example, greeting gestures or action gestures may have the same handshape but different motion patterns. Detecting these motion differences accurately is challenging.

3. Location Differences

Certain gestures use the same handshape but are performed at **different positions relative to the body**, such as near the face, chest, or head. Vision-based systems must accurately detect spatial positioning to identify the correct sign.

4. Orientation Variations

The meaning of a sign can change depending on the **orientation of the palm or fingers**, such as inward, outward, upward, or downward. Small orientation changes can be difficult for recognition algorithms to detect.

5. Non-Manual Signals

Some signs depend on **facial expressions, head movements, or body posture** to convey meaning. Detecting these non-manual signals is difficult because most systems primarily focus on hand gestures.

6. Similarity Between Alphabet and Number Signs

Certain alphabet fingerspelling gestures and number gestures have similar handshapes. Recognition systems must identify subtle differences in finger positions to distinguish between them.

7. Environmental Factors

Real-world video data introduces additional challenges such as:

- lighting variations
- complex backgrounds
- occlusion between hands
- different camera angles

8. Variation Between Signers

Different individuals may perform the same sign with **different speeds, hand sizes, or styles**, making it difficult to create a generalized recognition model.

- fast signing vs slow signing
- different hand sizes
- slightly different gesture styles